

Edu Genie-AI

AI-Powered Learning Assistant

Naan Mudhalvan Project

Name: Abirami Priyadharshini.P

Team member : Kavi mozhi .S

Semester: III Semester

Project Domain: Generative AI & Education

Project Description:

Edu Genie-AI is an intelligent web application that helps students learn topics, ask questions, create summaries, generate quizzes, and follow personalized learning paths using Artificial Intelligence.

Introduction

- Education is becoming more digital and interactive.
- Students often need quick explanations and study materials.
- Traditional learning methods may not provide personalized support for every student.
- EduGenie-AI uses Generative AI to provide instant learning assistance.
- It acts as a digital study companion for students.

Main Purpose

To make learning **simple, interactive, personalized, and accessible** using Artificial Intelligence.

Problem Statement

Students may face difficulties such as:

- Understanding complex topics.
- Finding simple explanations.
- Preparing quick study notes.
- Creating practice questions.
- Planning what to learn step by step.

Objectives

- Provide instant answers to student questions.
- Explain difficult topics in simple language.
- Generate short and useful summaries.
- Create topic-based quizzes.
- Provide structured learning paths.

- Improve independent learning.

Technologies Used

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Python
- Flask / FastAPI

Artificial Intelligence

- Google Gemini API
- Generative AI

Configuration

- .env environment variables
- Local development server

Working Flow

User → Web Interface → Backend API → Gemini AI → Response → User

Main Features

Key Features of EduGenie-AI

1. Ask a Question

Students can ask questions and receive AI-generated answers.

2. Explain a Topic

Complex topics are explained using simple sections.

3. Summarize

Long information is converted into short and useful notes.

4. Generate Quiz

AI creates multiple-choice questions for practice.

5. Learning Path

Creates a step-by-step roadmap for learning a topic

Explain Topic & Summarize

Learning Content Generation

Explain a Topic

The system provides:

- **Definition** – Basic meaning of the topic.
- **How It Works** – Main working concepts.

- **Example** – Simple practical example.
- **Key Points** – Important points for revision.

Summarize

The system:

- Understands the given topic.
- Identifies important information.
- Removes unnecessary details.
- Produces a short and easy-to-understand summary.

This helps students learn and revise quickly.

Slide 7 – Quiz & Learning Path

Quiz Generation

EduGenie-AI automatically creates quizzes based on the selected topic.

- Multiple-choice questions
- Four options for each question
- Correct answer provided
- Unique questions
- Unique answer options
- Useful for self-assessment

Learning Path

The system creates a structured learning roadmap:

Basics → Core Concepts → Practice → Advanced Topics → Project

This helps students learn systematically.

System Working & Error Handling

How the System Works

Step 1: Student enters a topic or question.

Step 2: Frontend sends the request to the backend.

Step 3: Backend processes the request.

Step 4: The request is sent to the Generative AI model.

Step 5: AI generates the required content.

Step 6: Backend validates the response.

Step 7: The result is displayed on the webpage.

Error Handling

- Input validation
- API error handling
- Response validation

- Demo Mode / fallback mechanism
- Clear error messages

Advantages & Future Scope

Advantages

- Easy to use
- Fast responses
- Personalized learning support
- Useful for different subjects
- Helps with revision
- Generates interactive quizzes
- Provides structured learning guidance

Future Scope

- User login and authentication
- Student progress tracking
- Personalized dashboards
- PDF export
- Multi-language support
- Voice-based learning
- More advanced learning recommendations

Conclusion

EduGenie-AI is an AI-powered educational assistant designed to support students in their learning journey.

It combines:

- **Artificial Intelligence**
- **Web Technology**
- **Interactive Learning**
- **Automated Content Generation**

The project helps students **understand topics, get answers, revise information, practice through quizzes, and follow structured learning paths.**

Thank You